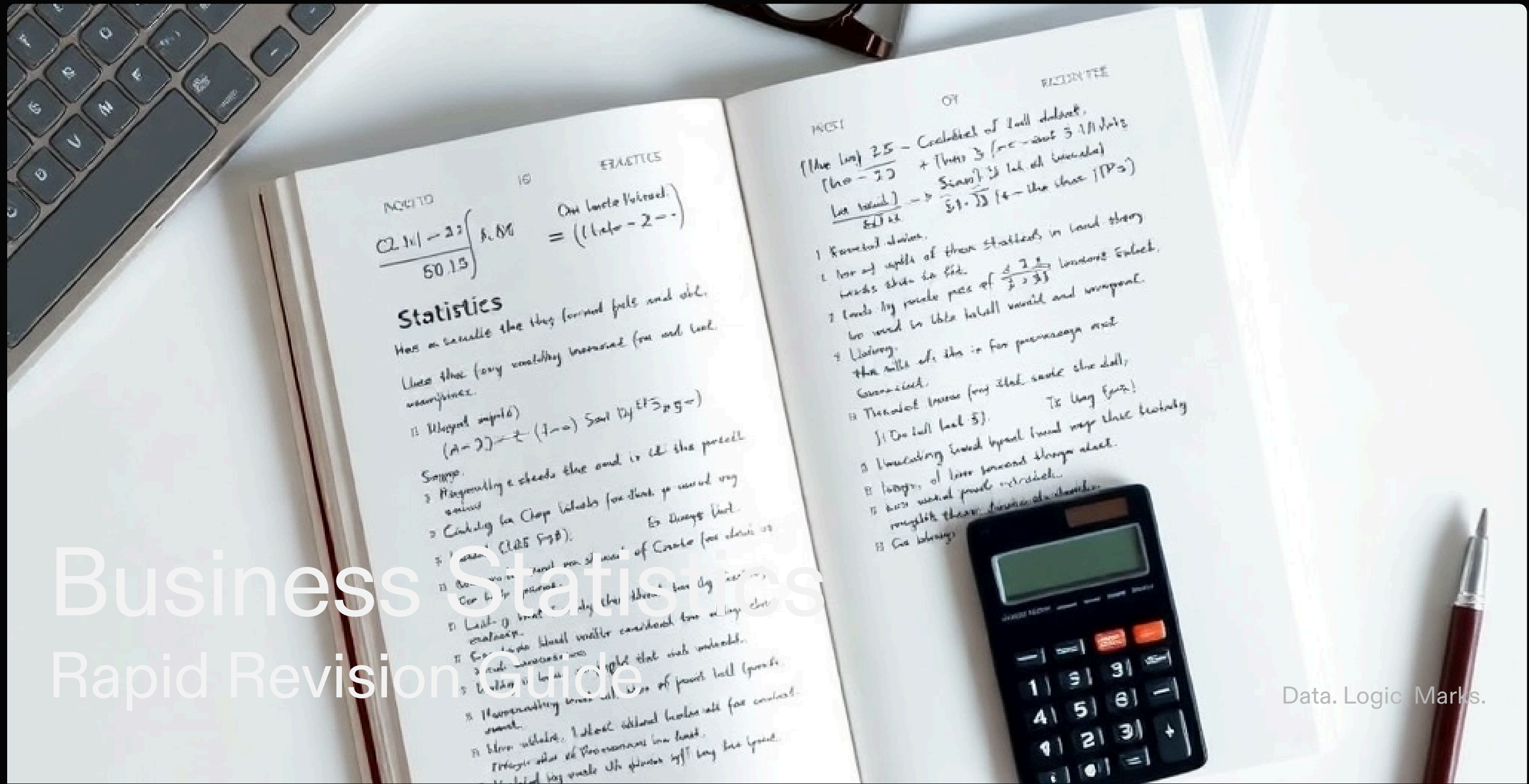
Business Statistics: Premium Rapid Revision Guide



Business Statistics

Rapid Revision Guide

Data. Logic. Marks.

Measures of Central Tendency

Deep Dive · Business Statistics

Core Formulas

01 Mean — Ungrouped Data

Mean = $\Sigma X / n$

02 Mean — Grouped Data

Mean = $\Sigma(fX) / \Sigma f$

03 Median — Grouped Data

Median = $L + [(N/2 - cf) / f] \times h$

04 Mode — Grouped Data

Mode = $L + [(f_1 - f_0) / (2f_1 - f_0 - f_2)] \times h$

When to Use

- **Mean** — Symmetric distributions, no extreme outliers
- **Median** — Skewed data or when outliers are present
- **Mode** — Categorical data or identifying most frequent value

Solved Examples

Average Monthly Sales

Sales figures (₹000s): 40, 50, 60, 70, 80
Mean = $\Sigma X / n$ = 300 / 5 = ₹60,000

↳ The business averages ₹60K in monthly sales — useful for budget planning.

Median Employee Salary

Grouped salary data: L=30K, N/2=25, cf=18, f=14, h=10
Median = 30 + [(25−18)/14] × 10 = ₹35,000

↳ Median is preferred here — high executive salaries would skew the mean.

Modal Product Sold

Modal class: L=20, f_1=18, f_0=10, f_2=12, h=5
Mode = 20 + [(18−10)/(36−10−12)] × 5 = 22.86 units

↳ Identifies the most demanded product — guides inventory stocking decisions.

Formula Key

- ΣX — Sum of all values
- n — Number of observations
- f — Frequency of modal/median class
- Σf — Total frequency
- L — Lower boundary of modal/median class
- N — Total frequency (for median)
- cf — Cumulative frequency before median class
- h — Class width (interval)
- f_1 — Frequency of modal class
- f_0 — Frequency of class before modal class
- f_2 — Frequency of class after modal class

Always state which measure you're using and why. Examiner marks are awarded for justification — not just the correct number!

📌 Exam Tip

Measures of Dispersion

Core Formulas

Range
Range = Highest − Lowest

Variance (Population)
$\sigma^2 = \Sigma(X - \mu)^2 / N$

Standard Deviation
$\sigma = \sqrt{\text{Variance}}$

Coefficient of Variation
CV = (SD / Mean) × 100

Why SD is Most Reliable

SD uses **every value** in the dataset — unlike Range, which only looks at extremes. It measures average deviation from the mean in the **original units**, making it intuitive and comparable.

Interpreting CV

CV expresses SD as a **%** of the mean, allowing comparison across datasets with different scales or units. A **lower CV** = more consistent. A **higher CV** = more variability.

Business Example: Comparing Sales Teams

Team A — Mean: ₹80,000 | SD: ₹8,000
CV = (8,000 / 80,000) × 100 = 10%

Team B — Mean: ₹60,000 | SD: ₹9,000
CV = (9,000 / 60,000) × 100 = 15%

Conclusion: Team A has a lower CV (10%), meaning their sales performance is **more consistent** despite a higher average. Team B shows more variability — useful for assessing reliability, not just output.

Use CV whenever means differ — it's the only fair comparison of spread.

Correlation & Regression

★ High-Priority Exam Topic

Karl Pearson's Coefficient (r)
$r = \Sigma[(X - \bar{X})(Y - \bar{Y})] / \sqrt{[\Sigma(X - \bar{X})^2 \times \Sigma(Y - \bar{Y})^2]}$

Range: $-1 \leq r \leq +1$

Interpretation:

- $r = +1$ → Perfect positive correlation
- $r = -1$ → Perfect negative correlation
- $r = 0$ → No correlation
- $|r| > 0.75$ → Strong correlation
- $|r| < 0.5$ → Weak correlation

Business Use: Sales vs. Advertising Spend analysis — measures how closely ad budget drives revenue.

Regression Equation (Y on X)
$Y - \bar{Y} = r\,(\sigma y / \sigma x)(X - \bar{X})$

Key Terms:

- $\sigma y / \sigma x$ — ratio of standard deviations (regression coefficient)
- $\bar{X}, \bar{Y}$ — means of X and Y
- r — Pearson's correlation

Solved Example 1: Given $r = 0.8$, $\sigma y = 10$, $\sigma x = 5$, $\bar{X} = 20$, $\bar{Y} = 40$. Find Y when $X = 25$.
$Y - 40 = 0.8 \times (10/5) \times (25 - 20)$
$Y = 40 + 8 = 48$

Solved Example 2: Given $r = 0.6$, $\sigma y = 6$, $\sigma x = 4$, $\bar{X} = 15$, $\bar{Y} = 30$. Find Y when $X = 19$.
$Y - 30 = 0.6 \times (6/4) \times (19 - 15)$
$Y = 30 + 3.6 = 33.6$

Probability

Core Formulas

Basic Probability
P(A) = Favorable Outcomes / Total Outcomes

Addition Rule
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Multiplication Rule (independent)
$P(A \cap B) = P(A) \times P(B)$

Conditional Probability
$P(A|B) = P(A \cap B) / P(B)$
Probability of A *given* B has occurred.

Bayes' Theorem
$P(A|B) = P(B|A) \times P(A) / P(B)$
Reverses conditional probability to update beliefs.

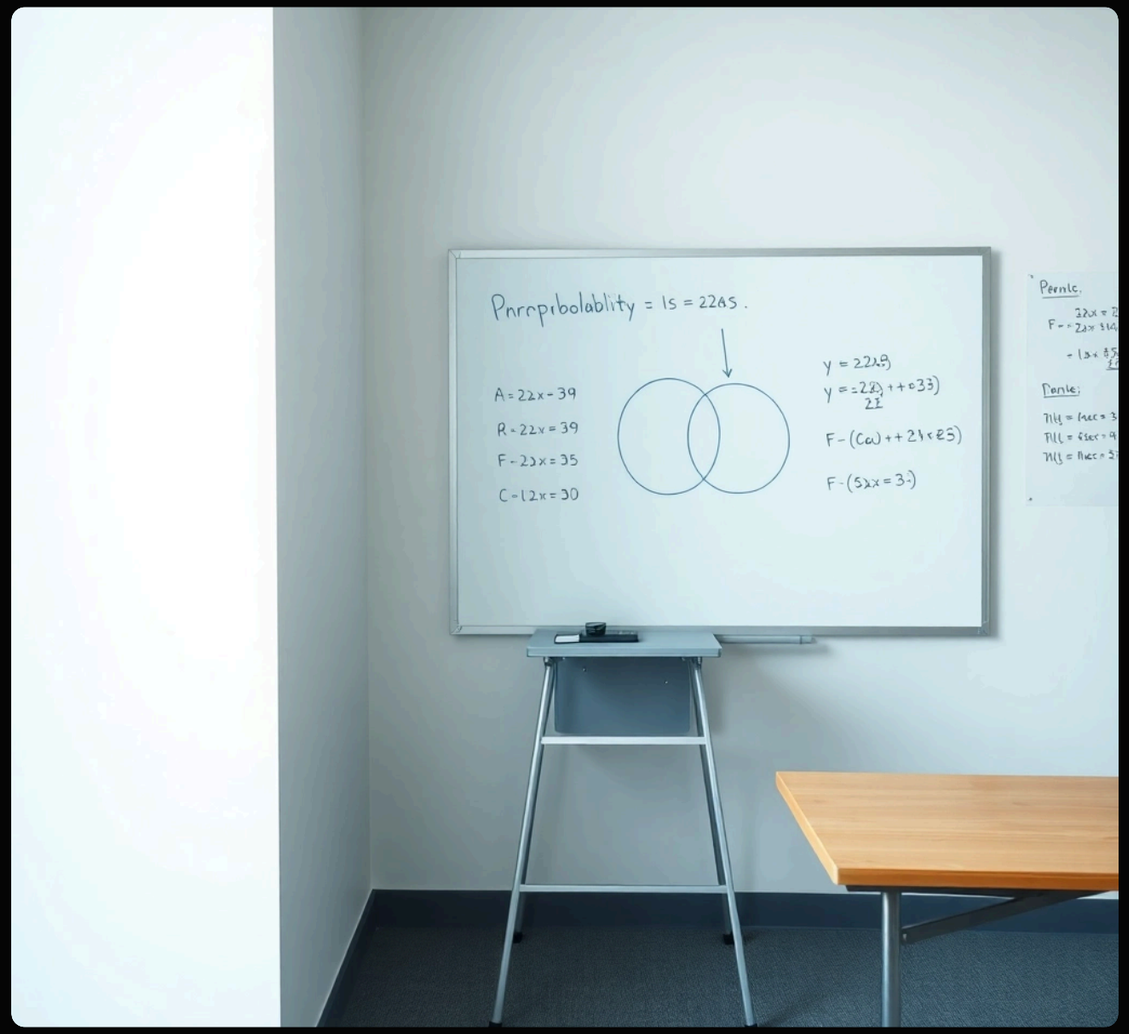
Solved Examples

1. Defect Rate
A batch has 5 defective items out of 50. P(defect) = 5/50 = **0.10**

2. Two Independent Tests
P(pass both) = 0.9 × 0.8 = **0.72**

3. Quality Testing (Addition)
P(A) = 0.4, P(B) = 0.3, $P(A \cap B)$ = 0.1
$P(A \cup B)$ = 0.4 + 0.3 − 0.1 = **0.60**

4. Bayes' (Defect Detection)
Machine flags defect 95% of time. False positive = 2%. P(defective) = 1%.
P(defect | flagged) = **≈ 32.5%**



Probability Distributions

DEEP DIVE · PROBABILITY

Binomial Distribution
Used when there are a fixed number of independent trials, each with two outcomes (success/failure).

Formula:
$P(X = r) = {}^{n}C_{r} \times p^{r} \times (1-p)^{n-r}$

Mean = np | Variance = np(1−p)

Example 1: A batch has 10 items; defect probability = 0.2.
Find P(exactly 2 defectives).
$P(X=2) = {}^{10}C_{2} \times 0.2^{2} \times 0.8^{8} \approx 0.302$

Example 2: 5 sales calls, success rate = 40%.
Find P(at least 1 sale).
$P(X \geq 1) = 1 - P(X=0) = 1 - 0.6^{5} \approx 0.922$

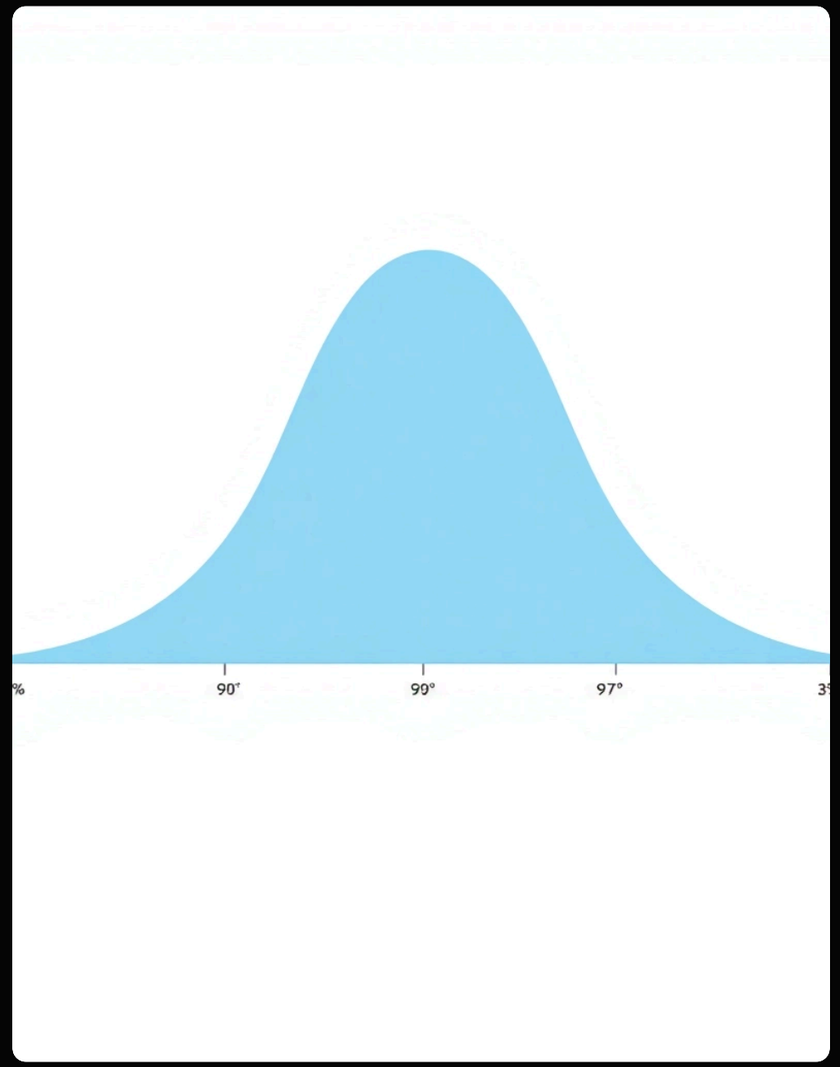
Normal Distribution
A symmetric, bell-shaped curve defined by mean (μ) and standard deviation (σ).

Standardisation (Z-score):
$Z = (X - \mu) / \sigma$

68−95−99.7 Rule:
68% of values lie within ±1σ
95% within ±2σ
99.7% within ±3σ

Example 3: Employee scores: μ = 70, σ = 10.
Find P(score > 85).
$Z = (85-70)/10 = 1.5$
$P(Z > 1.5) = 1 - 0.9332 = 0.0668$

Use standard normal table to look up Φ(Z) and compute required probabilities.



Index Numbers

Deep Dive · Formulas & Application

Three Core Index Formulas

Simple Price Index

$SPI = (P_1 / P_0) \times 100$

Measures price change of a single commodity relative to base year.

Laspeyres Index

$L = \Sigma(P_1Q_0) / \Sigma(P_0Q_0) \times 100$

Uses base year quantities (Q_0) as weights. Tends to overstate inflation.

Paasche Index

$P = \Sigma(P_1Q_1) / \Sigma(P_0Q_1) \times 100$

Uses current year quantities (Q_1) as weights. Tends to understate inflation.

CPI (Consumer Price Index)

A Laspeyres-type index tracking price changes of a fixed basket of goods. Used to measure cost of living and inflation over time.

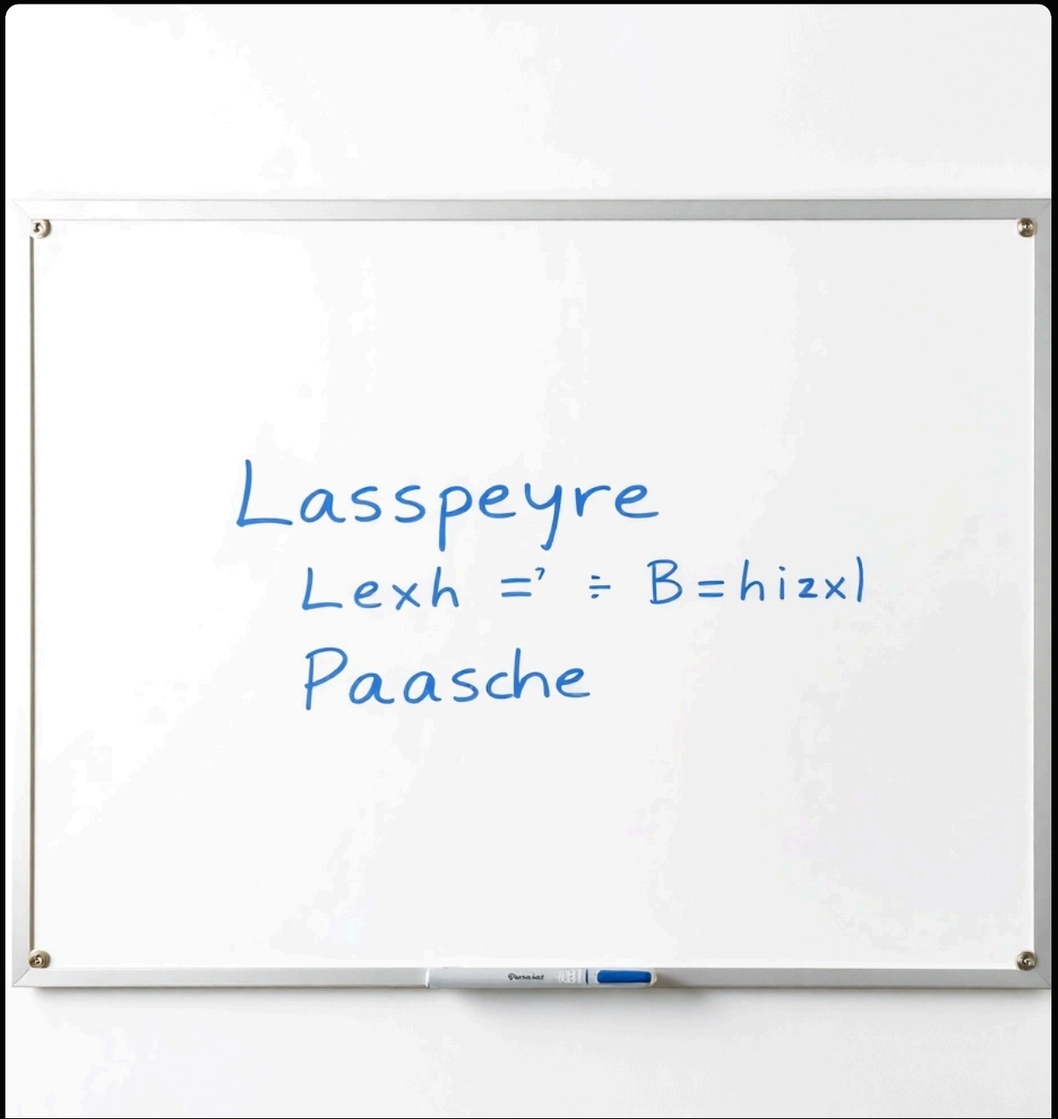
Limitations of Index Numbers

- Base year choice can bias results
- Fixed basket ignores substitution effects
- Ignores quality changes in goods over time
- Regional & income group differences are not captured

Solved Example — Laspeyres vs. Paasche

Two goods: Rice & Wheat

- P_0: Rice = ₹10, Wheat = ₹8
- P_1: Rice = ₹14, Wheat = ₹10
- Q_0: Rice = 50, Wheat = 30
- Q_1: Rice = 40, Wheat = 35



Time Series Analysis

1. Trend Analysis
Identifies the long-term direction (upward, downward, or stationary) of data over time. Used to forecast future performance by observing underlying patterns.

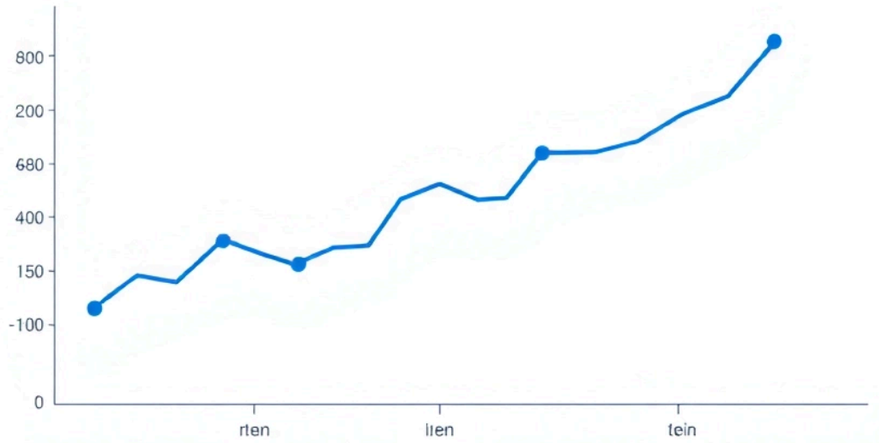
2. Moving Average Method
Smooths out short-term fluctuations to reveal the underlying trend.
Formula: $MA = \Sigma Y / n$

📊 Example: Monthly Revenue
Monthly revenue (₹000s): 40, 45, 50, 48, 52, 55
3-month MA (Month 2): $(40+45+50)/3 = 45$
3-month MA (Month 3): $(45+50+48)/3 = 47.7$

3. Seasonal Variation
Measures recurring fluctuations within a year due to seasons or cycles. Calculated by comparing actual values to trend values.
Seasonal Index = $(\text{Actual} / \text{Trend}) \times 100$

4. Least Squares Method
Fits a straight-line trend to time series data.
$Y = a + bX$
Where: $a = \Sigma Y/n$ | $b = \Sigma XY/\Sigma X^2$

📊 Example: Quarterly Sales
Given $\Sigma Y = 200$, $\Sigma XY = 60$, $\Sigma X^2 = 10$, $n = 4$
$a = 200/4 = 50$
$b = 60/10 = 6$
Trend line: $Y = 50 + 6X$



Quick tricks & exam strategy

01 **SD shortcut formula method:** Use the computational formula $\sigma^2 = (\Sigma x^2/n) - \bar{x}^2$ to save time — no need to find deviations from the mean individually. Faster and less error-prone under exam pressure.

02 **Correlation sign trick:** Before computing r, scan the data — if both variables increase together, r is positive; if one rises while the other falls, r is negative. Confirm direction before crunching numbers.

03 **Z-score table quick lookup:** Always identify whether the question asks for area to the left or right of z. Sketch a quick bell curve to avoid reading the table in the wrong direction.

04 **CV comparison method:** When comparing variability across different units or means, jump straight to CV = $(\sigma/\bar{x}) \times 100$. Skip recalculating SD unless asked — the ratio is all the examiner needs.

05 **Calculator usage strategy:** Use memory functions to store intermediate values (e.g., Σx^2, $\bar{x}$) and avoid re-keying. Practice the STAT mode for grouped data — it's the single biggest time-saver in numerical questions.

06 **Common probability mistakes:** Don't confuse P(A and B) with P(A) × P(B) — the multiplication rule only applies to independent events. Always check independence or use the conditional formula when in doubt.

Rapid Revision Formula Sheet

All key formulas — condensed for quick reference

CENTRAL TENDENCY

Mean (Ungrouped): $\bar{x} = \Sigma x / n$
Mean (Grouped): $\bar{x} = \Sigma fx / \Sigma f$
Median (Grouped): $M = L + [(n/2 - F) / f] \times h$
Mode (Grouped): $Mo = L + [\Delta_1 / (\Delta_1 + \Delta_2)] \times h$

DISPERSION

Variance (Pop.): $\sigma^2 = \Sigma f(x - \bar{x})^2 / \Sigma f$
Std. Deviation: $\sigma = \sqrt{[\Sigma f(x - \bar{x})^2 / \Sigma f]}$
Coeff. of Variation: $CV = (\sigma / \bar{x}) \times 100$

CORRELATION & REGRESSION

Pearson's r: $r = \Sigma(x - \bar{x})(y - \bar{y}) / \sqrt{[\Sigma(x-\bar{x})^2 \times \Sigma(y-\bar{y})^2]}$
Regression Line: $Y = a + bX$
$b = \Sigma(x - \bar{x})(y - \bar{y}) / \Sigma(x - \bar{x})^2$
$a = \bar{y} - b\,\bar{x}$

PROBABILITY DISTRIBUTIONS

Binomial: $P(X=r) = {}^nC_r \times p^r \times (1-p)^{n-r}$
Mean: $\mu = np$ SD: $\sigma = \sqrt{np(1-p)}$
Normal (Z-score): $Z = (X - \mu) / \sigma$

INDEX NUMBERS

Simple Price Index: $P_{01} = (P_1 / P_0) \times 100$
Laspeyres Index: $L = [\Sigma(P_1 \times Q_0) / \Sigma(P_0 \times Q_0)] \times 100$
Paasche Index: $Pa = [\Sigma(P_1 \times Q_1) / \Sigma(P_0 \times Q_1)] \times 100$

TIME SERIES — LEAST SQUARES

Trend Line: $Y = a + bT$
$b = \Sigma TY / \Sigma T^2$ (when $\Sigma T = 0$)
$a = \Sigma Y / n$

Where: L = lower class boundary, h = class width,
f = frequency, F = cumulative freq. before median class